

Microservices Benefits - A Detailed Insight

Microservices Benefits . . . Self-sufficiency (1)

- **Every microservice is an independent node used to build an app.**
- The decentralized approach helps develop self-sufficient applications that can work effectively by combining different technologies.
- **Each microservice is designed to help achieve certain business goals.** Therefore, developers can easily upgrade and update components without affecting the overall software.
- **When a new challenge appears, a business can easily develop and add a new microservice or upgrade the existing one to solve a new problem.**
- For instance, an online store owner can connect the following third-party tools using API integrations or HTTP/HTTPS requests:
 - an accounting system, or a payment gateway, or
 - an inventory management system, or
 - a customer portal, or a web analytics service
- **All the services can work independently and be maintained separately, which is one of the main benefits of microservices.**

Microservices Benefits ... Technical variety (2)

- Applications that are built using the monolithic architecture use the defined tech stack. Developers have to use selected set of technologies and adjust them to build new solutions.
- Oppositely, microservices offer the opportunity to use various technologies to achieve business goals. Microservices brings outstanding flexibility in the use of technologies.
- Every microservice can be developed using a set of different technologies and prog. languages to get best solutions possible. Businesses can choose technologies to use based on their access to tech talents or problems needed to solve.
- Selected popular programming languages and their best uses:
 - Java, GoLang — web and mobile applications, embedded systems
 - Python — machine learning and artificial intelligence
 - PHP — server-side web development and command-line scripting
 - SQL — data analysis and data mining
 - Ruby — back-end web and data processing
- Any programming language can be used to develop microservices to take the most advantage of every technology.

Microservices Benefits . . . Scalability (3)

- The ability to scale up an application fast is one of the most important features of the microservices architecture.
- Most businesses value outstanding flexibility offered by microservices, and as it uses a distinctive technology and database, it can be scaled up independently.
- Also, you can start small by using minimum required amount of resources for every microservice to keep your operating costs low. When load on one microservice increases, its database and computing power can be expanded separately.
- The dispersed architecture foresees the ability to upgrade your software by adding new microservices. For instance, if a business needs an alternative payment gateway, it can be easily connected and added to the checkout.
- In case additional functionality is needed to achieve business goals, software engineers can develop it to expand the number of services offered by a company.
- There is no limit to the number of microservices that can be connected. For example, the Netflix web application is divided into more than 700 microservices.

Microservices Benefits . . . High Sustainability (4)

- The use of microservices helps build highly sustainable applications.
- Microservices are loosely connected, so they work as independent parts.
- If a microservice stops working because of technical issues, the app's functionality is poorly affected.
- Moreover, other microservice will likely not be affected by a node's outage.
- Ability to allocate and fix errors is one of the top benefits of microservices.
- For instance, if a payment gateway installed stops working, users can explore the site, add goods to the cart, and even make purchases online if an alternative payment solution is available.

Microservices Benefits . . . Continuous Delivery (5)

- Continuous delivery is a part of CI/CD, a method that helps automate product development and release processes to increase the overall performance of teams.
- Since microservices work autonomously, developers can easily establish a continuous integration/continuous delivery environment.
- They can upgrade certain services without disabling the entire application.
- Software engineers can release new features effectively, keeping the maximum uptime of an application.

Microservices Benefits . . . Easy Changes Implementation (6)

- The decentralized architecture of microservice applications helps developers implement minor updates seamlessly.
- Software engineers can temporarily disable a microservice to implement minor changes, making a small piece of functionality unavailable for users.
- The app will remain unaffected because there is no need to redeploy the entire system to update some features when using the microservice architecture.

Microservices Benefits . . . Simplified Onboarding (7)

- Since the entire app is divided into small microservices that have distinctive tech stacks, the onboarding process is fast.
- New software engineers don't need to explore the source code of a huge monolithic application.
- Vice versa, they have to inspect a small microservice to start working on a new project.
- Microservices help draw new developers and onboard them faster, optimizing the overall product development time.

Microservices Benefits Summary



Self-sufficiency

Every microservice helps get certain functionality or achieve a distinctive business goal.



Technical variety

Different tech stacks can be selected for every microservice to achieve the best possible result using various technologies.



Scalability

The diversified architecture can be easily scaled up by adding new microservices or increasing specific resources.



High sustainability

Since microservices work independently, it increases the overall system reliability and helps localize errors effectively.



Continuous delivery

The architecture allows updating specific components without disabling the entire app.



Easy changes implementation

New features can be delivered by updating distinctive microservices. There is no need to release significant software updates.



Simplified onboarding

New developers can start working on a new microservice fast. They don't need to examine all the functionality of a system.

Microservices Key Use Cases

Due to the many benefits of the microservices architecture, it is widely applied to develop software solutions in different industries. Top five microservices use cases are the following.

■ Outdated Applications

- The use of microservice applications helps software engineers update legacy applications that can't help achieve business goals.
- They can break a monolith into microservices and apply new technologies to update separated parts of an application.

■ Apps with Complex Logic

- Applications like social media platforms and e-commerce web apps have a lot of components in their architectures.
- The development of applications with complex logic is one of the foremost microservices use cases.
- The microservice architecture foresees the opportunity to develop many various features using distinctive technologies and connect them.

Microservices Key Use Cases ... cont'd

■ Data-Heavy Applications

- Applications that have to process a lot of data fetched from different resources usually have a lot of dependencies.
- Microservices help use resources wisely, reducing the complexity of an app's architecture.

■ Apps that Process Real-Time Data

- Amount of data needed to be processed in real-time & stored may change drastically.
- Applications built using the microservice architecture can allocate the required amount of data storage and computing power to selected services upon demand.

■ Highly-Resilient Applications

- Apps built for industries like healthcare or finances should maintain 100% uptime and store data securely.
- With the help of the microservice architecture, developers can create highly-resilient and secure apps. Even when one or more microservices are down, the entire application can keep working, providing access to the rest of the functionality.

Statistics Summary on Microservices Adoption

- Many companies widely adopt microservices.
- According to an industry survey report, more than 81% of companies of different sizes already use microservices.
- Microsoft has conducted a comprehensive survey on the use of microservices. The presented results state that the top five microservices architecture use cases by app types are as follows.
 - Data analytics — 45%
 - Database applications — 41%
 - Customer relationship management — 38%
 - Finance — 34%
 - HR applications — 31%
- Roughly 92% of the microservice architecture adopters are satisfied with the chosen solution.

Microservices architecture - Ifs and Buts

- Microservices have a lot of benefits that businesses value.
- However, many companies start small with monolithic apps because releasing a monolith is easier and faster. Hence, microservices aren't the best solution for all companies.
- Best practice is to explore the most popular use scenarios when a business needs to use the microservice architecture.
- Microservices architecture is the right choice if you:
 - expect your startup to grow at a high pace
 - are not satisfied with product development time
 - want to use various technologies
 - can afford to employ several teams of developers
 - strive to develop a highly reliable and failure-proof application
 - want to be capable of modernizing your app fast

Microservices Implementation Stats

MICROSERVICES IMPLEMENTATION STATISTICS



81.5%
of companies already
use microservices

Already use



17.5%
of companies plan
to switch to the
microservice
architecture

Plan to adopt



1%
of companies will not
adopt microservices

Will not adopt



92%
of businesses reported
a successful switch to
microservice
architecture



84%
of architecture adopters
will keep investing in
microservices
development



28%
of respondents
highlighted the
improved application
quality and performance
as the key benefit of
microservices



84%
of companies reported
that microservices
adoption helped
improve teams
collaboration



87%
of business owners
stated that
microservices adoption
expenses are worth it



26%
of companies noticed
security improvements